

Elective course (Term 7)- Sept 2025

01.119/99.503: Nanoelectronics and Technology

Homework 2 –Due date 21 Oct, 9PM

1. Draw the SOI-FinFET structure and describe how current conduction mechanism is different from planar MOSFET. Briefly explain the reason for the improvement of I_{on} and I_{off} characteristics in FinFET compared to planar MOSFET.
2. With the help of a block diagram highlight and explain the different interconnect level in CMOS ICs.
3. How does interconnect scaling affect the reliability and failure of interconnects.
4. Compare the fundamental deposition mechanisms of Physical Vapor Deposition (PVD), Chemical Vapor Deposition (CVD), and Atomic Layer Deposition (ALD).
→ Highlight differences in source material phase, reaction control, and film growth.
Also include the schematic of the deposition equipments.
5. (a) Briefly explain why polysilicon is not used as a gate electrode material in High-k/Metal Gate (HK/MG) CMOS technology.
(b) Describe the Gate last process in High-k/Metal Gate (HK/MG) CMOS technology explaining the reason with respect to Gate first process in Poly-Si/SiO₂ CMOS technology.
6. (a) Explain why diffusion process of doping is less used compared to ion implantation in advanced CMOS technology, even though it is a simpler process.
(b) Compare diffusion and ion implantation in terms of:
 - Dopant control precision
 - Junction depth
 - Process temperature
 - Crystal damage
 - Equipment complexity

7. Using appropriate photolithography steps, obtain the final pattern on the Silicon wafer as shown in Fig. 1, below. Sketch the detailed steps followed during the whole photolithography process and name the type of photolithography used.

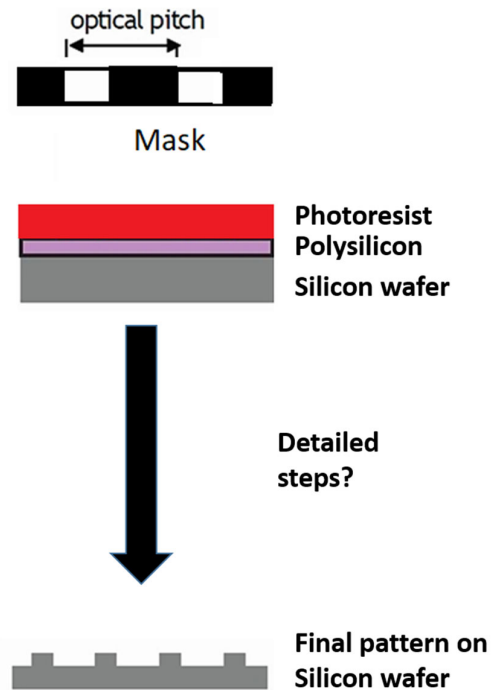


Fig. 1

8. Explain Extreme Ultraviolet (EUV) lithography procedure with the schematic of EUV system. Briefly discuss the challenges of using EUV lithography.